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# Cold chain optimisation models: A systematic literature review

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ABSTRACT

'Cold chains' are specialised supply chains that help preserve the quality of temperature-sensitive products, from procurement to fulfilment. They are vital for essential goods, including perishable food and medicine. Due to their refrigeration requirements, cold chains incur higher operational costs, energy consumption, and emissions than traditional supply chains. Additionally, they pose a risk of product decay or wastage. Despite the extensive research on cold chains, there is a need for a comprehensive review of optimisation models to consolidate existing knowledge and identify areas for further investigation. As such, cold chain optimisation models may deliver substantial commercial, environmental and social benefits, warranting a state-of-the-art overview.

We employ pertinent keywords to retrieve articles published up to and including 2023 from Scopus, followed by content analysis to select 234 publications from 122 peer-reviewed journals. First, we synthesise optimisation models and solution methods addressing collaboration, strategic network design, storage design, tactical planning and distribution planning. Next, we distil nine components, often unique to cold chains, prevalent in these models. Then, we establish multiple research trends and gaps (related to stakeholders, methodology, technology and novel approaches) in this burgeoning field, which will interest people exploring optimisation opportunities in cold chains. Finally, we point to future research avenues for academics, provide actionable learnings for firms' decision-makers, and outline governmental aspects for policymakers.

## 1. Introduction

Refrigerated supply chains, commonly known as 'cold chains', prevent exposure of goods to ambient temperatures. They provide chilling and freezing capabilities in operations — ranging from procurement to demand fulfilment — with cost and complexity higher than regular supply chains (Smith & Sparks, 2003). Berk (2018) classifies chilling into cold chill for meat at just above 0 °C, medium chill for dairy products up to 5 °C, and exotic chill for fruit and vegetables at 10–15 °C, while freezing uses typically −18 °C, but could be well below −70 °C for some vaccines. As a result, cold chains ensure perishables (such as fruit and vegetables, dairy products, meat, flowers, vaccines, medicines, and certain chemicals) attain a shelf life to meet market needs, satisfy health and safety regulations and retain their physical and chemical attributes.

The cold chain (CC) market had a revenue of USD 242 billion, some 5 % of the global supply chain (SC) market, in 2021, with an expected annual growth rate more than three times that of the global logistics market in the next decade (Grand View Research, 2022; IMARC, 2022). Although most perishables are food or medical products deemed essential, in a world with 924 million severely food-insecure (FAO et al., 2022), over 17 % of food production goes to waste (United Nations Environment Programme, 2021). CCs could help reduce some of this loss resulting from the decay of food products based on intrinsic and extrinsic factors (Cohen & Saguy, 1985; Kong & Singh, 2016). However, CCs need higher financial investment and demand more energy for cooling – contributing to greenhouse gas emissions. In addition, since fresh produce packaging lowers volumetric efficiency (leading to higher volume for the same shipment weight), it increases cooling needs in storage and transportation (Carson & East, 2018). Therefore, striking the right balance between the costs and benefits of CC capabilities is crucial. Long a challenge for operations management (Delen et al., 2011) and only ever as cold as their warmest link, CCs, if optimised, promise considerable commercial and social benefits.

SCs comprise a network of organisations involved in processes that deliver value to customers via products and services (Mentzer et al., 2001). Within SCs, perishables such as fresh produce need a different strategy to tackle product loss owing to deterioration (Blackburn & Scudder, 2009). CCs are specialised SCs that help preserve the quality of

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temperature-sensitive products from procurement to fulfilment. The role of CCs is crucial in managing the temperature profile of perishables over time — using refrigeration during pre-cooling, storage, transport and display (Mercier et al., 2017). The interdependencies between cooling effort, product decay, shelf-life and price, managed via collaboration among SC partners, are unique to CCs. Our work addresses this niche SC discipline.

Literature review papers address diverse topics of CC performance (Shashi et al., 2018), interactions between partners (Fan et al., 2020), sustainability (Vrat et al., 2018), technology adoption (Chaudhuri et al., 2018), and more recently, vehicle routing (Awad et al., 2021) and cooling kinetics (Ambaw et al., 2021). The CC for vaccines has been the focus of other reviews (Hanson et al., 2017; Lemmens et al., 2016; Matthias et al., 2007) and may evolve as the world handles complex temperature constraints applicable to some vaccines. Despite increasing use and sophistication in practice and a rich and rapidly expanding repository of underlying articles from the young field of optimisation models used in CCs over the past two decades, an extensive search found no literature review on this specialist CC topic. Existing reviews often focus on isolated aspects, leaving a gap in understanding the holistic optimisation of cold chains. We aim to address this conspicuous gap by systematically reviewing CC optimisation models across all SC functions. Our assessment of the state-of-the-art literature, synthesising practical applications of CC optimisation models (using different solution methods), would serve an audience exploring optimisation opportunities within CCs or extending SC optimisation models to CCs. An outline of future research directions should interest people engaged in modelling supply chain operations within academia, industry, and policymaking.

The paper proceeds as follows. *Section 2* describes the literature selection methodology — a descriptive summary of selected papers is presented in *Section 3*. The contributions from these articles are synthesised as CC optimisation models in *Section 4*, followed by an outline of critical model components of CCs in *Section 5*. *Section 6* highlights research gaps with a roadmap for future research. We offer concluding remarks in *Section 7*.

## 2. Methodology and motivation

To capture, condense and synthesise the body of knowledge in the field of cold chain (CC) optimisation models, we adapted the six-step systematic literature review process of Durach et al. (2017) to supply chain (SC) management and the niche field of optimisation modelling in the CC. This study comprehensively reviews optimisation models specific to cold chains, a rapidly growing and essential area within supply chain management. By identifying usage trends, model components, and future research directions, this review aims to guide researchers, practitioners, and policymakers in enhancing the efficiency and sustainability of cold chains. Table 1 outlines the steps and how we apply them, starting with research questions.

Our preference for this six-step process over other techniques, such as PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) (Page et al., 2021) is that the Durach et al. (2017) process is specifically tailored to the complexities of supply chain research. It allows for a comprehensive and iterative search strategy, crucial for capturing the diverse and evolving nature of optimisation models in cold chains.

For the literature search, we classify 'cold chain' more rigorously here as any SC implementing temperature control using refrigeration during various stages, such as production, storage, transportation or retail. We used Scopus, a well-established research database of peer-reviewed literature, to identify relevant articles (Harzing & Alakangas, 2016). Scopus was chosen for its extensive coverage of high-quality journals across various disciplines, including supply chain management and optimisation. This database's comprehensive indexing and advanced search capabilities ensure we capture a wide range of relevant studies, making it ideal for conducting a systematic literature review on cold chain optimisation models.

**Table 1**
Literature review steps (adapted from Durach et al. (2017)).

| Step | Our description |
|---|---|
| 1. Define research question(s) | What are the building blocks of CC models, and how are they used to optimise different parts of this rapidly growing field of SC? What trends and gaps point to future research in this domain? |
| 2. Determine the required characteristics of primary studies | Primary studies published in English journals up to and including 2023 (with no limit on the start date) and available in a research database as of the end of 30th November 2024. |
| 3. Retrieve sample of potentially relevant literature | Define search terms iteratively using the snowballing technique to cover all relevant scenarios. |
| 4. Select pertinent literature | Scan the title, abstract and paper content to ensure a relevant contribution to the research questions. |
| 5. Synthesise literature | Discern critical attributes related to optimisation models based on a detailed article study. Use an abductive approach proposed by Seuring et al. (2021): deductive method to map cold chain optimisation models to logistics theory proposed by Simchi-Levi et al. (2005) and inductive method to identify optimisation components in cold chains. |
| 6. Report the results | Consolidate the current body of knowledge – first by optimisation models in various SC functions (in *Section 4*), then by CC modelling components (in *Section 5*), followed by a descriptive summary (in *Section 3*) and suggested future research directions (in *Section 6*). |

Following the recommendations by Saenz and Koufteros (2015) for keywords, we defined our search terms to encompass optimisation models within CCs to shortlist articles for further screening. To begin with, 'optim*' and 'model*' capture optimisation models and variants of these terms. We searched 'cold chain' and its synonyms (identified by snowballing) and their variants, including indirect references to temperature control within SC, using 'logistics' as a proxy for SC, and limited our scope to English-language journal articles. Placing no filter on the publication start year allowed us to map the evolution of this field in which no previous reviews exist. Our search strategy may exclude references to optimisation without a modelling component and temperature control in SCs involving heating (such as fast-food delivery) or a narrow focus on a business function (such as production in controlled environments) without referencing CCs or SCs.

We show the scope of this review in Fig. 1, starting with papers representing the union of CC optimisation models [A] and other SC optimisation models addressing some form of temperature control [B].

The criteria and keywords used in the Scopus search are outlined in

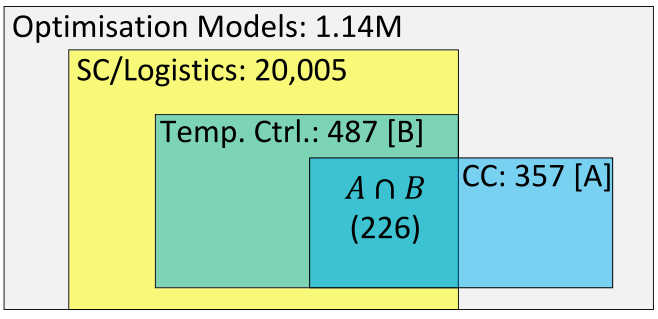



**Fig. 1.** Scope of the review A ∪ B (618).

```
TITLE-ABS-KEY (                                  # Search on Title, Abstract or Keyword
   (optim* AND model*) AND                       # Optimisation models
      (((supply-chain OR {logistics}) AND        # Supply Chain/Logistics with
        (cold OR refrigerated OR temperature)) OR # temperature control
      (cold-chain OR cool-chain OR chill-chain))) # Cold Chain (and synonyms)
SRCTYPE(j) LANGUAGE('English')                   # Journal articles in English
PUBYEAR < 2024                                   # Published until 2023
```

**Fig. 2.** Anatomy of the Scopus Search.

Fig. 2.

Relevant articles were chosen via title/abstract review and a detailed text screening. To include a paper, we looked for *optimisation* of CCs using an analytical or a linear programming model, *simulation-based optimisation* using a process model, or *regression-based optimisation* using experimental/empirical data. However, we excluded articles that did not address the role of temperature, directly or indirectly, in these models. Fig. 3 shows our progressive filtering process, starting with 618 publications representing the abovementioned overlap. Articles eventually used in this review were published from 2001 to 2023 (available in Scopus on 30th November 2024); we found no papers before 2001 despite having no filter on publication year.

A descriptive summary of these articles is provided in *Section 3*. Using a deductive approach, *Section 4* presents a detailed analysis of CC optimisation models in the selected literature grouped by their purpose, followed by a curated list of model components, identified using an inductive approach, in *Section 5*. Additional details about individual publications are available in the Supplement.

## 3. Descriptive statistics

This section summarises the publication landscape and critical attributes of the cold chain optimisation model publications. Besides a descriptive summary of the publications' metadata, we analyse their key features relevant to cold chains.

The 234 articles forming the basis of this review came from 122 journals, half of them sourced from just 22. We use the Herfindahl-Hirschman Index (HHI), typically applied to assess market competitiveness (Laine, 2016), to indicate the concentration of publications among journals. While an HHI of 10,000 represents a monopoly and 100 represents a highly competitive landscape, the value of 184 in our case indicates a highly fragmented landscape (as a benchmark, the ten review papers referenced in this article have an average of 421 and a standard deviation of 180 – see Supplement for details). Just eight journals contribute five or more items: *Sustainability* (18), *Computers & Industrial Engineering* (12); *Energies, IEEE Access,* and *Mathematical Problems in Engineering* (6 each); *International Journal of Production Economics, Transportation Research Part D: Transport and Environment,* and *Journal of Cleaner Production* (5 each).

Starting with intermittent articles from 2001 to 2015, we observe a notable growth in publications in this field in recent years, as illustrated in Fig. 4. Around 79 % of the articles appeared since 2019, with 58 % emerging since 2021. Despite a slight dip in 2023, the last four years indicate a growing importance of CCs in world trade.

There are 3.7 authors on average per article and only 10 single-author publications. Accorsi R., Liu S., Manzini R., Shi Y., Tao F., Wu D and Zhang C. are the most prolific authors (considering the number of articles and weighting contribution equally across all authors). Although potentially affected by our English language criterion, over 60 % of the publications have authors from Asia (China alone accounts for 80 % of these articles), followed by Europe (16 %) and the Americas (4 %). Only 17 % of the multi-author publications have co-authors across geographical regions. Our author affiliation analysis shows that while government agencies are research partners in 5 % of all papers, commercial organisations are rarely involved (<1%). Also, we note that around 60 % of the research is funded (by academic, national and international institutions), and Chinese institutions fund 67 % of such

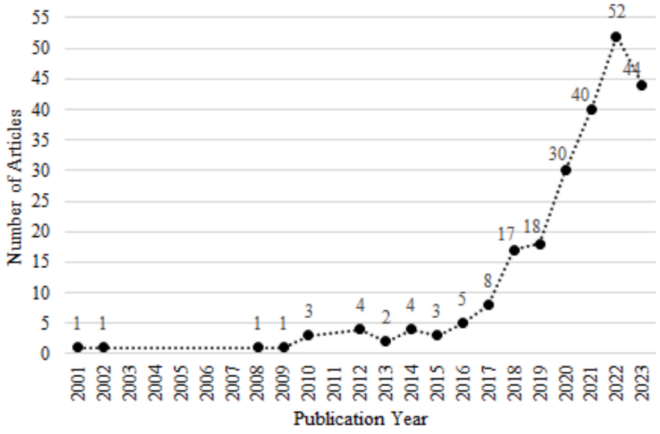



**Fig. 4.** Number of articles by year.

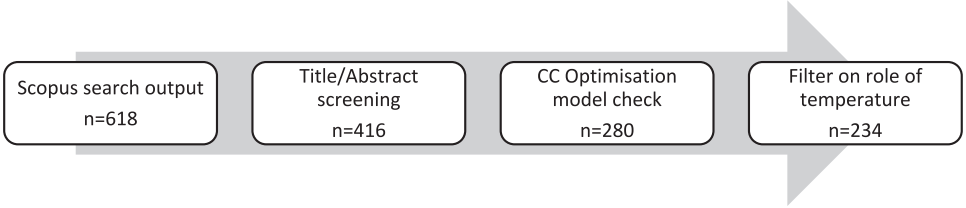



**Fig. 3.** Filtering process.

papers.

The h-index (or Hirsch index) evaluates the research output by an individual or a journal by the number of publications and their citation counts (Hirsch, 2005; Hodge & Lacasse, 2011). Schubert (2008) suggests an approach to assess the h-index of a single publication based on the number and citation count of articles referencing it. Adapting this to the context of review papers, we consider the h-index of the list of articles included in the review to indicate their overall influence. Our set of 234 journal articles has an h-index of 45 (at least 45 articles have more than 45 citations each). As a benchmark, the ten review papers referenced here have an average h-index of 44 with a standard deviation of 19 – see Supplement for details. The reviewed articles have an average of 30 citations (as per Scopus), with 14 papers having 100 + citations each and 30 having 50–99 each.

## 4. Cold chain optimisation models

This section captures and categorises, from the reviewed articles, the range of optimisation models prevalent in CCs. The model objectives are wide-ranging, including cost, profit, shelf life, travel time, service quality, and emissions. We use a deductive approach to align CC optimisation models with the SC Logistics theory proposed by Simchi-Levi et al. (2005). Here, we cover a range of models in CCs: strategic models that address SC collaboration and network design; models for designing storage assets related to packaging, transportation and warehousing; tactical models for production planning, storage and ordering; and operational models for distribution planning and monitoring. We also discuss the modelling approaches and solution methods adopted in the relevant papers. While many articles model more than one facet of CCs (see Supplement for details), a global optimisation model remains elusive.

### 4.1. Supply chain collaboration

SCs involve multiple entities, including suppliers, manufacturers, distributors, wholesalers, retailers, and logistics service providers (LSPs), strategically managing a distribution channel. SC collaboration entails two or more of these parties coordinating their decisions to generate higher overall channel profit than would accrue from optimising such decisions at the individual organisation level. In CCs, such coordination includes decisions regarding assets and operations related to temperature-controlled transportation and storage.

Strategic decisions about CC investments need coordination between producers, wholesalers and retailers. For a wholesaler making investments in the CC, information sharing with the producer via an incentive scheme can determine prices at each level to maximise profit for both parties (Cai et al., 2010). On a similar note, a retailer can incentivise the producer to invest in CC processes by adopting a bonus contract (C. Liu et al., 2021) or a revenue-sharing contract (Bai et al., 2019; Xu et al., 2021) to optimise the overall profit for all parties involved in the e-commerce delivery of fresh products. Other drivers for CC investment noted in the literature are: reducing food safety risk (Wang & Zhao, 2021), reducing emissions from transportation and warehousing (Ma et al., 2020; Ma et al., 2021), and protecting market goodwill (C. Liu et al., 2021) on account of increasing environmental awareness (Ma et al., 2022). While the government may not actively participate in such contracts, they can use subsidies to encourage CC investment and foster coordination among players to enhance profits (Xu & Cai, 2020). For instance, the government can incentivise the adoption of refrigerated electric trucks by subsidising manufacturers (to invest in research) and LSPs (to encourage purchases) (Zhao et al., 2022). Game theory-based models are commonly used to represent these interactions.

Another area of coordination is operational, covering distribution and fulfilment. For instance, an LSP can consolidate routes serving multiple distribution centres (DCs) and customer demand nodes in a joint distribution model, reducing costs and improving service levels (Y. Wang et al., 2021). The option of self-pickup by buyers is incorporated in such a model by Li and Shi (2023). As an alternative to joint distribution, vendor-managed inventory (VMI) can help lower network costs by sharing information (Stellingwerf, Laporte, et al., 2018) or assets like vehicles and storage (Q. B. Yu et al., 2023) across parties. In the case of fulfilment, credit terms set by a company with its suppliers and customers play a role in the optimal allocation of orders and demand by contributing a revenue/cost term in the form of interest earned/paid (Otrodi et al., 2019). While Bai and Wang (2022) explore 'put options' in supply contracts, Ghosh et al. (2022) assess the interaction between advance payment to suppliers to secure quality delivery and part-payment from retailers to secure orders to utilise the cold stage effectively. The collaboration models discussed above are developed as analytical expressions and solved using ordinary differential equations (ODE) to minimise operating costs or maximise channel profits.

Cai et al. (2010) provide an approach to model profit as a function of demand, price and cold chain investment ($\tau$): $\pi(p, q, \tau|\varepsilon) = p \bullet E[\min(D(p, \theta(\tau)), q \bullet m(\tau))] - (c(\tau) + w) \bullet q$, where demand (D) depends on price (p) and freshness level ($\theta$); supply (q) received at cost (w) is limited to the survival fraction (m); and investment influences cost (c), survival and freshness level; and randomness is represented by $\varepsilon$.

### 4.2. Strategic network design

SC networks comprise nodes (locations for production, processing, packaging, storage, and point of sale) and arcs (routes for transporting products between these nodes). Overall logistics costs in such a network include fixed expenses for facilities and vehicles, variable expenses related to ordering, inventory holding and transportation, and CC-specific costs such as refrigeration, deterioration and waste disposal (Kaviyani-Charati et al., 2022; Sinha & Anand, 2020; Ullah et al., 2019; Wangsa et al., 2023). Optimising this cost objective determines the number and location of DCs that should be opened in the CC network (Li & Zhou, 2021). DC location determines construction costs (Dou et al., 2020), with S. Y. Zhang et al. (2021) including competing DCs and their existing flows in such a model. Wang et al. (2018) further incorporate the financial impact of carbon emissions related to storage and transport in the overall cost. For greenfield projects, the choice of DC location in network design depends on cooling costs, product deterioration costs (Ji et al., 2021), emissions (K. Li et al., 2022), and DC capacity (Jaigirdar et al., 2023) or demand fulfilment capability (Rahbari et al., 2022). These SC networks are usually modelled in the CC literature as Mixed-integer linear programming (MILP) models, coded using languages like AMPL or GAMS and solved using exact methods with commercial solvers such as CPLEX, Gurobi, and LINGO.

Some network models also address variability and options relevant to CCs. For instance, product freshness within the CC depends not only on refrigerated storage and transportation investment but on random factors like weather and delays, influencing the proportion of products reaching their destination at an acceptable quality level (Cai et al., 2010) or retaining freshness at the store (D. Zhang et al., 2023). Similarly, when designing distribution networks (say, for vaccines) in developing countries, many of which are hot or lack reliable electricity, an adequate number of cold storage devices at each echelon must be modelled as a constraint (Lim et al., 2019). Variability is generally handled by robust optimisation (Ji et al., 2021; Qu & Wu, 2021; Wei et al., 2023; F. Yang et al., 2022) or scenario linearisation techniques (Babagolzadeh et al., 2020). Since some fresh products, such as green peas, can be stored in a chilled or frozen state, the model could recommend either option based on the differential storage cost (Zanoni et al., 2019). A bi-level optimisation is suitable for modelling the government's role in CC: at a high level, the government optimises overall emissions and subsidies; at a lower level, each enterprise optimises its profit (S. Y. Zhang et al., 2019). When network design optimisation is combined with vehicle routing, *meta*-heuristics are preferred to exact methods (Li, 2022; Wang et al.,

2022).

A typical formulation of such an optimisation model is illustrated by S. Y. Zhang et al. (2021), where the upper-level objective function: $\min_x C_x(x, y(x))$ s.t. $H_x(x, y(x)) \leq 0$ reflects decisions made at the network level (x), followed by the lower-level decisions at the distribution level (y) influencing objective function: $\min_y C_y(x, y(x))$ s.t. $H_y(x, y(x)) \leq 0$. Upper-level decisions determine the location, number and size of DCs, while lower-level decisions respond to demand by allocating resources for delivery.

### 4.3. Storage design (package, box, container, and warehouse)

Primary packaging helps control the atmosphere around a product during storage, transport, and retail within the CC via adequate ventilation and temperature control. Since micro-perforations regulate humidity and atmospheric gases in fruit packages, Jalali et al. (2017) propose an analytical model based on permeability and heat transfer to simulate and guide perforation design. Similarly, packing material regulates heat exchange, emphasising the role of insulation. An optimal combination of beet pulp and date palm fibre with gypsum is proposed as a natural insulation medium by Farrokh et al. (2021). Anand et al. (2022) note a 30 % increase in storage duration by duplicating two layers of insulation plus coolant using an equal amount of material to determine an optimal packaging solution for transporting animal tissue in CCs. Since cooling involves complex inter-relationships that cannot be easily expressed as analytical expressions, these scenarios are modelled using CFD with simulation-based optimisation.

Secondary packaging (cartons) and tertiary packaging (pallets) are designed to facilitate handling and transportation. They also assist in uniformly cooling products in the CC (Gong et al., 2021). At this packaging level, cooling efficiency is influenced by the choice of open vs. closed top cartons (Wu et al., 2019) and the position of vents in cartons and pallets (Zhao et al., 2016) – this phenomenon can be evaluated using CFD. Another approach is employed by East and Smale (2008): the temperature profile of products in a small-volume shipment box is modelled based on convective and conductive heat transfer principles and compared with experimental data to help identify optimal box design. Such shipments may also use eutectic plates (filled with a solution typically frozen using off-peak electricity to provide cooling without energy later) for transporting products within a recommended temperature and humidity range (E. H. Liu et al., 2021). The choice of box size and the number and size of eutectic plates can be determined experimentally (Tsang et al., 2018). Finally, Quaternary packaging refers to refrigerated or insulated containers transporting products in the CC. Temperature distribution in such containers can be modelled using CFD to improve the cooling rate and uniformity based on the appropriate number and placement of doors (Zhang et al., 2018), the position and speed of chiller fans (B. Li et al., 2020), and thermal insulation material and thickness (He et al., 2022). Maiorino et al. (2022) integrate photovoltaic panels to design a hybrid vehicle for refrigerated transport to reduce emissions. Specific CFD software (like ANSYS) or simulation methods (like Monte Carlo) are commonly used for optimisation.

Energy modelling allows balancing the thermal loads with the cooling capacity to achieve an acceptable temperature change (Maiorino et al., 2022) during storage and transportation. For items with mass (m) and specific heat capacity (c), this is depicted as $mc\frac{\partial T}{\partial t} = Q_{Aux} + Q_{Defrost} + Q_{Door} + Q_{Wall} + Q_{Cabinet} - Q_{Ref}$, where Q is the thermal load generated by auxiliary components, defrost cycle, door usage and heat exchange through wall/cabinet compensated by refrigeration to limit change in temperature ($\partial T$) over time ($\partial t$).

Refrigerated warehouses require higher initial investment (fixed) and ongoing operating costs (variable) than regular warehouses due to additional cooling equipment and increased energy needs. Automated storage and retrieval systems (AS/RS) are widespread in such warehouses to limit workers' exposure to extreme temperatures. An optimal design of such AS/RS minimises the costs related to machinery and energy consumption of equipment and cooling systems (Meneghetti & Monti, 2015) for the operational capacity needed. Photovoltaic installation to reduce warehouse emissions has been integrated with this model by Meneghetti et al. (2018). A similar approach is presented by Wu et al. (2019), where solar energy is used in precooling at the warehouse instead of cooling during transportation. While thermodynamic principles can be used to model warehouse temperature and moisture control mechanisms (Park et al., 2018; Wang & Lu, 2022), a 'digital twin' approach is proposed by Hu et al. (2023). Bottlenecks in CC warehouse operations can be identified via discrete-event simulation software (Zhu et al., 2014). Finally, a bi-level objective model applicable to storage is proposed by Du et al. (2022): optimising location and routes at the upper level, followed by warehouse layout and order allocation at the lower level. Telle et al. (2022) extend this by including electric vehicles to optimise the overall energy needs of the CC.

### 4.4. Tactical planning (procurement, production, storage, and fulfilment)

Multi-period tactical planning in CCs covers procurement, production, warehousing, and fulfilment. Procurement also determines ordering policies that help manage raw material inventories in SCs. In CCs, such policies should include the perishability of stock over time (Zanoni & Zavanella, 2012), the cooling cost for storage (Bozorgi et al., 2014), and quality issues noted on inspection (Alamri et al., 2016). In addition, trade credit periods may influence the optimal replenishment frequency and order quantity of perishables (Ouyang et al., 2010).

A production plan prescribing actions needed in each period based on demand, supply and resource availability is usually formulated as a MILP yielding exact solutions for straightforward cases (Pasha et al., 2021; Wei et al., 2019; Zheng et al., 2017; Zhou & Zong, 2021). For instance, a multi-period farming activity is modelled based on changes in the harvesting phase over time (Aliano Filho et al., 2023), accounting for losses due to deterioration (Orjuela-Castro et al., 2021). The impact of climate patterns on crop yields and, thus, on commodity prices is included in such a model by Flores and Villalobos (2018) to determine optimal decisions for planting, harvesting and transportation across multiple regions. For CCs dealing with seasonal products, workforce variation is an area of concern – Chaabane et al. (2022) address this challenge by including labour stability as a social dimension in a multi-objective optimisation model when planning CC operations. These tactical models can handle uncertainty in production and demand using a rolling horizon plan to adjust for changes over time (Lejarza & Baldea, 2020).

For storage, planning involves allocating warehouse rooms (Keshavarz-Ghorbani & Pasandideh, 2021) and activating specific freezer units (Hariga et al., 2017) while accounting for the impact of room utilisation on cooling effectiveness (Marchi et al., 2020). These plans influence product quality at the retail level – this is noted for blueberries, where storage temperature determines mass loss and sweetness, as shown by regression over experimental observations (Casagrande et al., 2021). Since cold storage may lead to chilling injuries, Han et al. (2021) propose an intermittent warming approach for apples to achieve uniform temperature and moisture while reducing energy use. Adjusting storage capacity using rented cold stores based on actual harvest outcomes is explored by Mateo-Fornés et al. (2023). When dealing with multiple products, their allocation to different storage temperatures can be determined by balancing the storage cost with the loss of product quality (De Freitas Borghi et al., 2009). Finally, catering operations present an unconventional temperature-controlled storage need: meals can be held at either a low temperature or, breaking from strictly 'cold' chains, a high one to reduce food safety risk, a decision influenced by resource availability and energy consumption (Tufano et al., 2020).

On the demand side, fulfilment policies modelled on shelf life instead of traditional 'first-in-first-out' (FIFO) are more appropriate for CCs

(Koutsoumanis et al., 2002). Such shelf-life models can also set prices based on dynamic expiry dates for perishables (Tromp et al., 2012). Perishability also influences fulfilment planning at a retail level – closing a store during some day(s) of the week, applicable to many stores worldwide, increases product loss due to deterioration. Such losses are higher for products with random deterioration rates than those with fixed perishability (Chołodowicz & Orłowski, 2021). Furthermore, a retailer's replenishment strategy depends on the extent of quality loss vs quantity loss of specific products (Wang, 2022). Therefore, replenishment timelines for decaying items should be synchronised with their production schedule to optimise overall SC cost, as G. Chen et al. (2022) note for an orange juice manufacturer. Simulation-based optimisation is commonly used in these models.

Optimisation models typically integrate production–distribution planning (Pasha et al., 2021) to maximise quality at the lowest overall cost. The objective function for quality can be described as max $Z_1 = \sum_{meg} Y_{meg} \bullet Q_g$, where m is manufacturing facility, e is demand node, and g is the quality grade for the product. $Y_{meg}$ is the binary decision variable representing path selection and $Q_g$ is the quality level after production. The cost function is given by min $Z_2 = \sum_m Y_m \bullet c_m^{Op} + \sum_g w_g \bullet c_g^W + \sum_{mg} p_{mg} \bullet c_{mg}^{Pr} + \sum_{meg} Y_{meg} \bullet d_{meg} \bullet c_{meg}^{Tr}$, where unit cost parameters $c^{Op}, c^W, c^{Pr}, c^{Tr}$ represent operating, wastage, production and transportation. $Y_m$ indicates if a manufacturing facility is used, $w_g$ is the wastage for each grade, $p_{mg}$ is the production, and $d_{meg}$ is flow that caters to the demand.

### 4.5. Distribution planning (routing, delivering, rerouting)

Distribution entails allocating inventory to DCs, grouping customer deliveries, assigning vehicles for delivery, and planning vehicle routes. Many publications address this aspect of CC operations, covering vehicle routing and real-time adjustments. The Vehicle Routing Problem (VRP) is a generalisation of the classic Travelling Salesman Problem with multiple trips with one or more vehicles. Such a need for multiple trips results from constraints such as vehicle capacity (Peng, 2019; Wang & Yin, 2018; Xie & Li, 2019), weight limits (Jia, 2022), travel time limits (Jiao et al., 2021), variable speed during journey (Wu et al., 2023), urban speed limits (Shi et al., 2022), quality thresholds (Ma & Zhu, 2023) or delivery time preferences (Yan & Zheng, 2020). All these constraints are more restrictive in CCs, as explained further in this review. The variety of products and their temperature requirements drive the desire to enhance VRP models with multi-compartment vehicles or multiple depots (Fu et al., 2022; Li & Li, 2022) and an option to share vehicles and products (Ren et al., 2019). There is a detailed literature review of VRPs within CCs by Awad et al. (2021) and another of multi-compartment vehicles within SCs by Ostermeier et al. (2021). While the former addresses only transportation, quality and emissions, the latter offers broader coverage of multi-compartment vehicles and is not specific to CCs. The present review takes a more holistic view of VRP, treating it as part of the overall distribution process optimisation in CCs.

Last-mile delivery, accomplished via efficient vehicle routes, is a significant aspect of any distribution operation. In CC distribution, loss of product quality due to decay is dependent on travel time and can be modelled as a linear (Hsiao et al., 2018) or an exponential (Y. Li et al., 2019) relationship. Therefore, the travel duration from a depot to customers per prescribed time windows influences the quality of products delivered to individual nodes on the network (Hsiao et al., 2017; Liu et al., 2019). The Vehicle Routing Problem with Time Windows (VRPTW) and its variants form the core models for CC distribution. For instance, the concept of 'hard' (or preferred) and 'soft' (or acceptable) time windows allows delivery outside the preferred timings, with some penalty according to the extent of deviation (Liu et al., 2018). Furthermore, uncertainty in time windows can be modelled as a fuzzy set when determining optimal routes (Ali et al., 2021; Lian, 2021; Moghaddasi et al., 2023). An optimal path for a single vehicle considering penalty cost as an exponential function of satisfaction level is proposed by Tang (2022) and compared with Dijkstra's algorithm. Finally, the maximum travel time to the last delivery point can be limited per the quality loss threshold.

A typical objective function for optimal distribution is formulated as min $C_{Total} = C_{Labour} + C_{Travel} + C_{Idle} + C_{Cooling} + C_{Emissions} + C_{Penalty}$, where C represents the cumulative cost of each category. Penalty costs may be elaborated as: $C_{TimeWindowViolation} + C_{QualityLoss} + C_{Disposal} + C_{LostSale}$. In addition to vehicle routing constraints, customer preferences for delivery windows must be accommodated while ensuring the delivered products meet promised quality thresholds.

Since most VRP scenarios are non-trivial, solutions based on exact methods may be impractical. Instead, *meta*-heuristics, either search-based (W. Chen et al., 2023; Hübner & Ostermeier, 2019; Shao & Lu, 2023; R. Yu et al., 2023) or population-based (Hou et al., 2022; Ning et al., 2021; G. Zhang et al., 2023; Zhang & Wen, 2023; L. Y. Zhang et al., 2021), are frequently employed. In some cases, hybrid solutions, such as merging genetic algorithms with simulated annealing (Nakandala et al., 2016; Tao et al., 2023; Yin & Tian, 2022), hill-climbing (Fu et al., 2023) or K-short circuit (Liu & Zhang, 2023) are proposed. Variants of ant colony optimisation (Feng et al., 2023; Miao et al., 2023; Pande et al., 2022; Shu et al., 2021; D. Q. Wu, J. Y. Cui, et al., 2022; D. Q. Wu, Z. W. Zhu, et al., 2022) are also common. MATLAB has a range of built-in *meta*-heuristic methods and is the most referred-to software in the literature. In some scenarios, clustering techniques such as *k*-means, hierarchical or DBSCAN can group customer locations when allocating demand to a distribution centre (Liu & Zhang, 2022; Ren et al., 2019), determining vehicle routes (Y. Wang et al., 2021), reducing waste (Köseli et al., 2023), or prioritising specific demand nodes for vaccine distribution (Kochakkashani et al., 2023). Similar clustering methods apply to source nodes in the case of marine fishing and transportation (Yixuan et al., 2023).

Overall transport cost in a typical SC can be modelled as the sum of vehicle, driver, fuel, emission costs and penalties. CCs entail added costs of refrigeration, quality violation, and potential disposal. Refrigeration costs may be accounted for as additional fixed costs of vehicles, variable fuel costs dependent on external temperature (Chai, 2016), and resultant emissions (Yang et al., 2017). The impact of changes to the vehicle load (during a delivery) on cooling needs and fuel consumption can be included in the cost model (Stellingwerf, Kanellopoulos, et al., 2018), as can a correction for a road network with gradients (Leng et al., 2020). Road congestion can also be incorporated using empirical data by assigning an overhead cost based on region and time (Keming, 2015; Liu & Zhang, 2021; Qi & Hu, 2020). In addition, penalties may result from time window violations, quality threshold breaches, disposal and lost sales. As a result, a convex relationship between cost and vehicle speed is apparent in CCs: increased speed causes higher fuel consumption yet reduces expenses related to time window violation and deterioration. Y. Yang et al. (2022) address another source of penalty specific to a pandemic: the psychological impact of vaccine deliveries on drivers having to enter infectious regions.

Emissions in CC transportation result from increased fuel usage due to cooling requirements and refrigerant leakage from the cooling units. VRPTW for CCs can include emissions as an objective function or a constraint, depending on the carbon policies prescribed by regulatory bodies. Governments may consider four different approaches to control emissions (L. Li et al., 2019), which we describe here. The 'Carbon Cap' policy applies a limit, thus treating emissions as a constraint. A 'Carbon Cap Offset' requires a company to buy carbon credits to compensate for emissions above a limit, thus including the incremental cost in the objective function. A government may allocate carbon credits to a company in a 'Carbon Cap & Trade' scheme that allows it to sell/buy to/from the market to compensate for reduced/increased emissions. This policy incentivises companies to reduce costs and even profit from unused credits. The last option, a 'Carbon Tax', is relatively

straightforward: the company pays a tax based on the volume of emissions.

CCs influence some of the choices in a typical SC distribution network. Distributed products' volume and temperature specifications dictate the choice of vehicles with identical features vs. a combination of vehicle sizes and features (Zhu et al., 2021). When delivering multiple products with different temperature requirements (usually ambient, insulated, chilled, and frozen), using multi-compartment vehicles (MCVs) is the most suitable option (Frank et al., 2021) – they are more efficient than multiple deliveries utilising a variety of vehicles (Hsu et al., 2013; Jiang et al., 2021). Another choice is transportation mode (e.g., rail, truck or ship), which should depend on the temperature profile of the routes (Gallo et al., 2017). The climate patterns along these routes dictate the choice of a refrigerated vehicle (for a warmer region), with or without solar panels (Meneghetti et al., 2021), vs. a regular vehicle (for a cooler path) by balancing the costs related to distance, cooling and loss of quality (Accorsi et al., 2017).

Cargo ship routes carrying refrigerated goods can also be scheduled using VRP (Zhang et al., 2020; Z. Zhang et al., 2021) – here, the trade-off between wastage (due to the over-ripening of fruit) and fuel consumption based on ship speed is addressed by Fan et al. (2021). A trade-off between cost and customer satisfaction is also applicable in unimodal vs. multimodal transport for the export of refrigerated products (Zheng et al., 2022). Similarly, cost and flexibility in refrigerated vehicles can be considered simultaneously: one option is to provide the ability to switch between refrigerator and freezer modes in delivery vehicles (Al Theeb et al., 2020); another is to provide adjustable compartments that can be configured during loading to handle varying product volumes (Hübner & Ostermeier, 2019). Finally, when including emissions from transportation to optimise vehicle routes, the specific policy promoted by the government (carbon cap, cap and offset, cap and trade, or carbon tax) could influence both fleet selection and route (L. X. Li et al., 2019; Qin et al., 2019; Wang et al., 2017; L. Y. Zhang et al., 2019; B. L. Zhao et al., 2020; Z. Zhao et al., 2020). Since cost, service and emissions are frequently modelled as multiple objectives for VRP, a solution can be derived using appropriate weights (Chen, 2020) or trade-offs made using a Pareto front (K. Li et al., 2020; Mo et al., 2022).

Next, we consider distribution models that address the combined optimisation of multiple operations. For the last-mile distribution of groceries in CCs, a two-stage VRP with an initial delivery to transfer stations and final delivery to customer locations is more suitable (Z. Liu et al., 2021; Yan et al., 2020). Using electric vehicles (EVs) for grocery delivery in the urban context has been modelled by Z. X. Zhao et al. (2020) to include interim recharging activities at appropriate locations. Leyerer et al. (2020) further include the placement of multi-temperature parcel lockers in a distribution network to permit customer collection, along with delivery by electric cargo bikes. In addition, the loading sequence can be integrated with routing plans (Theophilus et al., 2021; Zheng et al., 2020), penalties for missed consignments (Rusdiansyah et al., 2023), and efficient unloading plans (Ostermeier et al., 2018). Furthermore, including the impact of traffic conditions in routing allows for a more realistic distribution plan addressing fleet requirements, cost, and emissions (Bai et al., 2022; Chen et al., 2021; Guo et al., 2022; L. S. Hu et al., 2021; L. J. Wu, Z. G. Chen, et al., 2022; A. Zhang, Y. Zhang, & Y. Liu, 2023). Combined optimisation of network design, inventory location, and vehicle routing is well suited for *meta*-heuristics (Wang et al., 2020) for large-scale problems or non-linear solvers such as BARON via free internet-based NEOS servers (Amaruchkul et al., 2022) for smaller ones.

SC operations involve real-time distribution plan adjustment, covering inventory at rest and in motion. Regularly monitoring temperature, humidity and light exposure using wireless sensors within a warehouse can help identify optimal pallet placements to maximise shelf life (De Venuto & Mezzina, 2018). Loisel et al. (2022) propose a neural network-based model to predict temperature across multiple zones within a pallet. Since transporting perishables over long distances may lead to an inconsistent decline in product quality, investing in a temperature monitoring solution facilitates decision-making (Tsang et al., 2021) at a relevant point on the path: continue the journey, divert the flow to a closer location, or abandon the shipment (White & Cheong, 2012). However, when demand exists for multiple product quality levels, Khan et al. (2020) argue that a compliance-based solution (rerouting or discarding on arrival) is cheaper than a monitoring-based one. For cases where CC deliveries must handle dynamic orders during distribution, Y. H. Chen et al. (2022) and Y. Chen et al. (2023) integrate order picking with delivery to handle demand surges, while L. J. Wu, L. Shi, et al. (2022) optimise fleet size to reduce order rejections. Disruptions during CC delivery are another area of concern: while Zulkefly et al. (2021) establish an optimal production recovery plan based on the impact of transport disruption, Hou et al. (2022) and Liu (2023) include the impact of traffic during deliveries for multi-compartment vehicles with a pre-optimisation approach followed by real-time adjustments. Unplanned road congestion may also be addressed by stopping the vehicle and delaying the delivery to reduce overall emissions (A. Zhang, Y. Zhang, et al., 2023). Shi and He (2018) propose a different approach to address the breakdown of a vehicle carrying vaccines: they explore sending an additional rescue vehicle instead of rerouting an existing vehicle to salvage the consignment.

## 5. Cold chain model components

Despite an extensive literature search, we found no cold chain optimisation theory, so in this section, we adopt the inductive approach Seuring et al. (2021) recommended. Based on a detailed analysis of optimisation models showcased in *Section 4*, we identify nine supply chain (SC) aspects unique to CCs, highlight their significance and provide representative examples. We discuss the components specific to CC optimisation models. We start with how food decay and heat/mass transfer, influenced by temperature, impact quality and, thus, shelf life, which governs pricing and demand. We then explore the effect of cooling dynamics on packaging decisions, fuel consumption, energy needs, and emissions in the overall temperature-controlled SC. While food chemistry and fluid dynamics are not strictly OR (Operations Research) topics, they are crucial to CC performance, in contrast to ambient SCs. We also evaluate the role of time windows (affecting stakeholder satisfaction) in the distribution of refrigerated goods. Finally, we investigate technology-based CC monitoring. Though some papers address more than one of these components (see Supplement for details), none proposes a grand model encompassing them all. Our purpose is to enable modellers to develop innovative solutions specific to their business situation.

### 5.1. *Microbial growth*

Microbial concentration, which influences the product quality of perishable goods, depends on temperature, humidity, and light exposure, as characterised by mathematical models (De Venuto & Mezzina, 2018). In addition, the extent of spoilage by specific microbes depends on temperature and atmospheric gases (Giannakourou et al., 2001) – microbial count (measured as colony-forming units) generally increases with temperature and $O_2$ concentration and reduces with $CO_2$ concentration. Most microbes grow above a specific high-temperature $T_H$, perish at a low-temperature $T_L$, and survive between $T_L$ and $T_H$. The recommended storage temperature for frozen food products is based on the common $T_L$ for microbes – typically −18 °C (Zhang & Chen, 2014).

Models for microbial growth are used to predict their concentration over time at a given temperature. Typically, their concentration $N_t$ at time t is modelled as $\frac{\partial N_t}{\partial t} = \text{Rate} \bullet N_t$, where the 'Rate' at temperature T is negative for $T < T_L$, 0 for $T_L \leq T \leq T_H$, and positive for $T > T_H$ (Mishra et al., 2016). Solving this equation facilitates the prediction of microbial concentration (above the threshold temperature $T_H$) over time with a

non-linear model: $N_t = N_0 \bullet e^{k\bullet(t-\lambda)}$, where $N_0$ represents the initial concentration, and $\lambda$ is the time lag in growth (Giannakourou et al., 2001). The impact of temperature T (the absolute temperature in Kelvin) on the coefficient k is modelled using two popular alternatives: Arrhenius' equation $k = k_0 \bullet e^{-\frac{E_a}{RT}}$, where $k_0$, R and $E_a$ are constants (Tromp et al., 2015); or Belehradek's equation $k = b^2 \bullet (T - T_{min})^2$, where $T_{min}$ is the temperature below which no microbial growth can occur, and b is a product-specific constant (Wang et al., 2019; Yifeng & Ruhe, 2013). Fig. 5a and 5b show the indicative microbial growth using these two models for hotter and colder scenarios. Unfortunately, the literature provides no specific guidance addressing the suitability of either model to specific situations.

Other models for microbial growth use a sigmoidal function that ranges from a minimum to a maximum value. A third model, commonly referred to as the four-parameter logistic model, has been applied to products such as cultured fish (Koutsoumanis et al., 2002) and is modelled as: $\log(N_t) = \log(N_{min} + \frac{N_{max}-N_{min}}{1+e^{-\mu_{max}(t-t_{1/2})}})$, where $N_{min}$ and $N_{max}$ represent minimum and maximum microbial concentration respectively while $\mu_{max}$ is the maximum growth rate (a function of temperature based on Arrhenius' equation) and $t_{1/2}$ is the time to reach half of the maximum concentration. Though not mentioned explicitly, one can infer that $t_{1/2}$ decreases with temperature. Finally, the modified Gompertz function: $\log(N_t) = A + C \bullet e^{-e^{-B(t-M)}}$ is the most suitable method for predicting microbial growth in fresh-cut potatoes (S. Zhao et al., 2022) and processed meat products such as pork chops (Tromp et al., 2012) or hamburger meat (Shi et al., 2010). In this fourth model, A represents minimum microbial concentration; C represents the difference between the maximum and minimum microbial concentration; and B is the maximum growth rate, which increases with temperature; and M is the time to reach maximum growth rate, which decreases with temperature. Fig. 5c and 5d show the indicative microbial growth using these models for hotter and colder scenarios.

### 5.2. Quality and shelf life

CCs attempt to manage product quality via temperature control to limit microbial growth – which influences product quality. Degradation in quality can be modelled as $\frac{\partial Q}{\partial t} = -k \bullet Q^n$, where $k,n > 0$, indicating a decay over time (Nakandala et al., 2016). The parameter n influences the reaction's order, with $n = 0$ representing constant decay (usually applied to fruit and vegetables) and $n = 1$ representing exponential decay (e.g., in the case of meat and fish) being the most common options. The degradation rate, k, is modelled using the familiar Arrhenius' equation (Ma et al., 2018). Thus, quality is defined as a function of temperature over time as $Q(T,t) = Q_0 - (k_0 \bullet e^{-\frac{E_a}{RT}}) \bullet t$ for zeroth order decay or $Q(T,t) = Q_0 \bullet e^{-(k_0 \bullet e^{-\frac{E_a}{RT}})\bullet t}$ for first-order decay, as shown in Fig. 6. These models explain the continuous inventory loss during storage and transportation (Tromp et al., 2015).

A standard measure of quality is shelf life, which reflects the time for product quality to become unacceptable to customers, regulators, and/or store management. Shelf life is modelled as a percentage change over time: $\Delta SL(\%) = \frac{100}{S_0} \bullet Q_{10}^{\frac{(T-T_0)}{10}}$, using $Q_{10}$ (the ageing factor based on the increase in deterioration for every 10 °C rise in temperature) and the known shelf life $S_0$ at reference temperature $T_0$ (Accorsi et al., 2018). An alternative is to use empirical data, as is done for fresh fish, to model shelf life reduction over time (t) at temperature (T) as $\Delta SL = k \bullet (1 + 0.1 \bullet T)^2 \bullet t$, where k is a constant (Alamri et al., 2016). However, in practice, shelf-life prediction at a given temperature is simplified to a linear model: $SL_T = SL_L - \frac{(T-T_L)}{(T_H-T_L)} \bullet (SL_L - SL_H)$, so that deterioration per unit of time can be computed as $1/SL_T$ (Hsiao et al., 2018). Here L and H represent lower and upper bounds.

### 5.3. Demand vs. Price and quality

The inter-relationship of price, demand, and quality drives revenue. In general, demand correlates negatively with price, described as $D_p = D_0 - k_p \bullet p$, where $k_p$ represents the price elasticity of demand. On the other hand, price correlates positively with quality, with a dampening effect that can be modelled as $p = p_0 \bullet (Q - Q_e)^n$, where Q represents the quality when the item reaches the customer, $Q_e$ represents the lower threshold (at expiry), $p_0$ is a constant, and $0 < n < 1$ describes the impact of quality loss on price (Shui & Li, 2020). Now, demand correlates positively with quality, and this interaction can be modelled as $D = D_p \bullet (\frac{Q}{Q_e} - 1)$. As an alternative, Cai et al. (2010) present a single-step multiplicative model: $D = D_0 \bullet q \bullet p^{-k}$, where $0 < q < 1$ represents the level of freshness, and k represents the price-elasticity factor. A more straightforward approach models demand as a linear function of price and quality: $D = D_0 - k_p \bullet p + k_Q \bullet Q$, where $k_Q$ represents sensitivity to quality (Yang et al., 2017), while a dynamic quality model is proposed by Xie et al. (2023). Demand is also influenced positively by goodwill (C. Liu et al., 2021) and credit period (Otrodi et al., 2019). Furthermore, when agricultural products are offered in both ordinary and CC forms at prices $p_O$ and $p_C$ respectively, the incremental demand for the ordinary variant is modelled as $D_O = k_p \bullet (p_C - p_O)$ (Xu & Cai, 2020).

### 5.4. Transpiration and respiration

Fruit and vegetables exhibit an increase in transpiration (release of moisture) and respiration (consumption of $O_2$ to release heat and $CO_2$) as temperature increases (Jalali et al., 2017). In addition, heat transfer

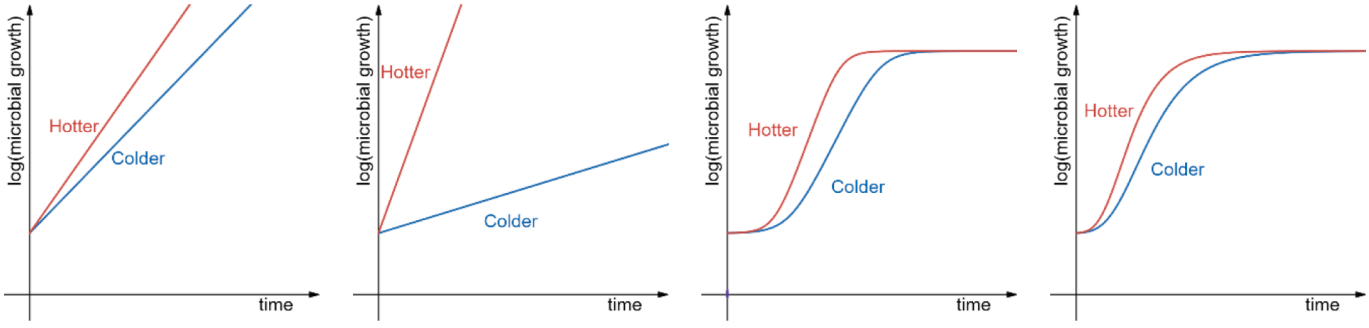



**Fig. 5.** Microbial growth (log value) over time and temperature.




**Fig. 6.** Quality over time and temperature (linear and exponential decay).

from the ambient atmosphere via packaging material impacts the atmosphere surrounding the object during storage (Jalali et al., 2020). The impact of ethylene emissions also needs to be considered for many fruits, as it creates a reinforcing loop of hastening decay (Chen et al., 2018). These interactions are depicted in Fig. 7. Some of these exchanges also lead to mass loss (Casagrande et al., 2023) and an increase in the sugar content of produce during storage (Casagrande et al., 2021; Shrivastava et al., 2023).

### 5.5. Dynamics of cooling

Packaging helps control the atmosphere surrounding products during storage and transportation. To model the temperature and airflow around the product within the CC, computational fluid dynamics (CFD) is preferred over expensive and time-consuming experiments (B. Li et al., 2020). In some cases, CFD is complemented with advanced experiments to build a temperature–time profile of individual products within a packaging arrangement to develop a virtual CC (Wu et al., 2019). These CFD simulations help evaluate cooling time, rate, and uniformity using the fundamental laws of conservation of mass, momentum, and energy (Nasser Eddine et al., 2022; Zhao et al., 2016). In addition to respiration (discussed above), a rise in temperature during storage also results from heat transfer via walls (which depends on the surface area, thickness, thermal conductivity, and temperature difference) and opening of doors for loading/unloading (which influences airflow) (Zhang et al., 2018). These temperature changes and relative humidity influence product maturity over time, which can be evaluated based on surface colour, as shown by Han et al. (2022) for tomatoes and Schudel et al. (2022) for strawberries. Shoji et al. (2022) model fruit via a 'digital twin' to assess the impact of cooling methods on shelf life at the retail stage. Similarly, Vera-García et al. (2022) model a refrigerated facility using 3D software to estimate energy use.

### 5.6. Fuel requirement

Vehicles in a CC network need thermal energy ($E_t$) for cooling requirements, and motive energy ($E_m$) for transportation. $E_m$ is a function of distance, total weight, the slope of the road, speed and vehicle-specific attributes (Stellingwerf, Kanellopoulos, et al., 2018). For refrigerated vehicles, the total weight would include compressors (at times with a dedicated engine) and insulation, and vehicle-specific attributes such as the expanded frontal surface area due to the refrigeration unit would affect air drag (Wang & Wen, 2020). The energy requirement for refrigeration is a function of the Coefficient of Performance (COP) of the unit, defined as heat absorbed per unit of energy supplies, formulated as $E_t = \frac{H}{COP}$, where H is the heat energy removed from the vehicle (Stellingwerf, Kanellopoulos, et al., 2018). This heat energy depends on travel time, the surface area of the cooling cabin (where products are stored in a refrigerated truck or trailer), the thermal conductivity of the cabin material, and the temperature difference between the cabin and the environment (Shui & Li, 2020).

Eventually, the energy need is converted to fuel usage using $F = \frac{E}{\eta \bullet E_f}$, where $E_f$ is the energy content per unit of fuel, and $\eta$ represents conversion efficiency. Since the load carried by the vehicle varies during distribution, a more straightforward approach is to approximate the load-dependent fuel consumption ($F_q$) using mileage during empty hauls ($F_0$) and with a full load ($F_Q$): $F_q = F_0 + \frac{(F_Q - F_0)}{Q} \bullet q$, where Q is the total capacity and q is the current load (L. X. Li et al., 2019; Ma et al., 2023; Wang et al., 2020). Besides cooling during travel, refrigerated vehicles also need extra cooling during unloading due to warmer air entering the cabin when the door opens (Chen, 2020; Liu et al., 2020).

### 5.7. Emissions

Sustainability is considered one of the key performance indicators of SCs (As'ad et al., 2020). Unlike regular SCs, CCs have higher energy needs due to cooling for storage and transportation. Besides electricity usage and exhaust gases, refrigerant leakage also contributes to emissions (Vu & Ko, 2023). Usually, a factor converts electricity or fuel consumption to $CO_2$ in kg (F. Li et al., 2022), potentially followed by a carbon credit price in $ per kg $CO_2$ (B. Hu et al., 2021). While $CO_2$ is the most measured exhaust in the case of transportation, other greenhouse gases such as $CH_4$ and $N_2O$ are also a cause for concern (Y. Li et al., 2019). In addition, the refrigerant loss can be modelled based on estimated annual leakage and applying a conversion factor to compute emissions (Stellingwerf, Kanellopoulos, et al., 2018). Abbas et al. (2023) provide an option to include other sustainable development goals within

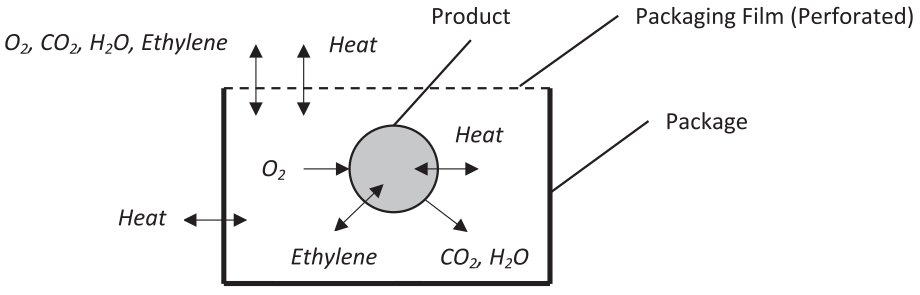



**Fig. 7.** Exchange of heat and gases.

the overall supply chain cost.

Total emissions can be modelled as $E_T = \sum_{ijkt} e \bullet \rho_{ijkt} \bullet x_{ijkt} \bullet d_{ij} + \sum_{it} g \bullet p \bullet (I_{it} - \frac{q_{it}}{2})$, where i and j represent nodes, k represents vehicles, and t is a period (L. X. Li et al., 2019). For transport, e is emissions per unit of fuel consumed, ρ is fuel consumption per unit (x) transported, and d is the distance between nodes. Similarly, for storage, g is emissions per unit of electricity, p is electricity used for storing a unit for a period, I is the initial inventory, and q is the demand (we assume a steady consumption of inventory).

### 5.8. Delivery time windows

In 'vehicle routing' parlance, the 'preferred' or 'hard' time window refers to a customer's ideal timeslot to receive deliveries (Brito et al., 2012), e.g., 8 am – 10 am. Customers may sometimes agree to 'shoulder periods', e.g., between 7 am – 8 am and 10 am – 12 noon, with diminished satisfaction. This broader range is called the 'acceptable' or 'soft' time window, where lower customer satisfaction results in a penalty (M. Wang et al., 2021). This penalty usually depends on the deviation from the preferred time (L. Chen et al., 2019; Yao et al., 2022) and may be nonlinear (Hsu et al., 2013; Yang & Tao, 2023) for some refrigerated products – arriving outside acceptable time windows is usually treated as a lost sale. Such penalty may be pro-rated in case of split-deliveries (B. Xu et al., 2023). Sometimes, the penalty rate may be proportional to the consignment value (J. Chen et al., 2019). If the delivery vehicle arrives before the acceptable time window, the driver may have to wait, during which time their refrigeration equipment stays active (Chen, 2020). However, if it arrives before the preferred, but within the acceptable, window, the trade-off between the early delivery penalty vs. waiting costs drives decisions.

The cost of penalties related to time windows can be formulated as:

$$C_{TW} = \begin{cases} p_{ls} \bullet q_i & if \quad a_i < E_i \\ p_e \bullet q_i \bullet (e_i - a_i) & if \quad E_i \le a_i \le e_i \\ 0 & if \quad e_i \le a_i \le l_i \\ p_l \bullet q_i \bullet (a_i - l_i)^{\gamma} & if \quad l_i \le a_i \le L_i \\ p_{ls} \bullet q_i & if \quad L_i < a_i \end{cases}$$

i: location with target delivery time between $e_i$ and $l_i$ of quantity $q_i$; $a_i$ is the arrival time; $p_{ls}$ is lost sale penalty; $p_e$ / $p_l$ are penalties for early/late delivery; γ adds non-linearity.

Timely delivery positively correlates with customer satisfaction: although customers are open to receiving deliveries within a time window for refrigerated products, their satisfaction within that time slot is higher when vehicles arrive earlier than later (Leng et al., 2020). On the other hand, driver satisfaction decreases with increased waiting time on arrival (Qiu et al., 2020). Therefore, a weighted combination of waiting time (which is inversely related to the overall satisfaction of customers and drivers) can be modelled by assigning a higher weight to a customer's waiting time. Furthermore, damage to product value may also influence customer satisfaction (Su et al., 2022).

### 5.9. Monitoring

Proper management of perishables involves monitoring the condition of goods throughout the SC (Alamri et al., 2016). Such monitoring consists of sensors and data loggers (Li et al., 2023; Shoji et al., 2022), along with technologies such as radio-frequency identification (RFID), global positioning systems (GPS), geographic information systems (GIS), and mobile data communication to analyse cooling performance during transportation (Chen, 2020; Z. Yang et al., 2022) and predict product temperature (Park et al., 2023). Furthermore, simple, inexpensive time–temperature indicators (TTIs) enable auditing the temperature history of products (Gaukler et al., 2023) – some even record data down to individual product level (Giannakourou et al., 2001; Giannoglou et al., 2014), typically using flexible sensors on packages (J. Xu et al., 2023). Maiyar et al. (2023) model distance thresholds to determine if monitoring is worthwhile. Oversight of CC products during storage is also pertinent – a wireless sensor network (WSN) can monitor warehouse temperature, humidity and light exposure (De Venuto & Mezzina, 2018), while Bankole et al. (2023) use predictive controllers for display cases storing vaccines. These technologies allow a range of preventive and corrective actions within the overall CC.

## 6. Research trends, gaps, and future directions

*Section 4* showed the diversity of optimisation models prevalent in cold chains (CCs), synthesised into different model components outlined in *Section 5*. Recent growth in peer-reviewed publications – over 85 % of the selected papers postdate 2017 – affirms the increasing popularity of this topic. During the last four years, articles appeared at an average rate of over ten per quarter. From our comprehensive literature review, we derive the immense potential of this niche field and identify fruitful research opportunities.

For instance, while the role of temperature in CCs is self-evidently crucial, optimisation models deal with it differently. Only 9 % of the articles consider it a decision variable prescribed by the model. The remaining papers deal with it as a constraint: 47 % concern products at a fixed temperature; 11 % address the need to hold products at multiple temperatures, usually in separate vehicle compartments or storage locations; and the last 33 % tackle variability in external temperature, where the temperature differential influences the product decay and the associated cooling effort. We posit that multi-temperature optimisation models will increase with the growing need for online grocery deliveries and the availability of multi-compartment delivery vehicles.

Solution validation strategies employed in the literature vary considerably. We have found four common approaches in the literature. First, numerical experiments involve testing the model with sample data, usually randomised within a business context, and developing scenarios using factorial design (for limited combinations), random design (for many variables), or Taguchi design. Second, standard databases are frequently used in VRPs – the Solomon dataset (Solomon, 1987) being the preferred option to benchmark solution performance (usually by tweaking the *meta*-heuristic parameters) and compare it to standard algorithms or heuristics. Third, reference data from an earlier paper, with appropriate modifications, is used in some cases to compare the proposed solution with the one in the referenced article to quantify improvements. Finally, company-specific data is used in many papers, with a few reporting benefits compared to existing policies. However, none of these discusses the benefits the company accrued after adopting the proposed solution.

58 % of the papers deal with a company case study, focusing on decision-making. Of these, 81 % anonymise the company name, and 9 % omit the location. Overall, 135 papers report case studies with their site details, with 96 instances from Asia (79 from China). Europe has 15 contributions (five from Italy), while the Americas has nine (four from the United States). This observation aligns with the institutional affiliation of the authors and the funding agencies noted earlier. We note that studies based in China are more likely to report the company name(s) in the publication.

The research articles address various product categories employing temperature-controlled supply chains (SCs). While some papers do not mention any specific product, others address multiple items. The Supplement details CC products with as much specificity as could be inferred from the articles.

This section explores the low-temperature storage and transportation needs of CCs, the subjective assessment of perishables' quality and recent market trends regarding consumer preferences and technology evolution. We suggest new research areas and potential improvements to current models and solution methods, grouped into four subsections. Stakeholders within CC operations and research are the focus of *subsection 6.1*. In *subsection 6.2*, we delve into solution methodology and

validation techniques. The role of engineering and IT is addressed in *subsection 6.3*. Finally, *subsection 6.4* suggests novel approaches to formulate business requirements for CC optimisation and publish research in this domain.

### 6.1. Stakeholders

The consumer perspective – what happens to a product during or after purchase – is missed in the optimisation models addressed in the reviewed literature. While quality is evaluated in terms of temperature, colour, and firmness, it ignores how consumers view freshness through visual appeal, aroma, texture, or taste. Ultimately, the threshold for quality depends on individual consumer preference, and models could include such variation. For example, the refrigeration capabilities available to consumers at home might determine their tolerance for quality (captured by surveys). Models could consider this aspect when choosing optimal product fulfilment, pricing, and disposal policies.

No studies recommend best practices for retailers to determine optimal pricing, ordering, and waste reduction policies specific to products requiring the CC – retailers are merely treated as 'sinks' while optimising product flow. For instance, if models for fulfilment planning integrate storage availability at the retailer, one may avoid full or partial rejection of delivery due to capacity issues. Furthermore, since most retailers deal with multiple suppliers, coordinating deliveries with other service providers as part of route planning may help improve the retailer's efficiency.

A few articles explore the government's role in setting emissions or subsidy policies. However, government can also actively enable stakeholder collaboration. For example, it can encourage asset sharing by distributing emission credits to a group of entities rather than individual agencies. Policies regarding investing in road infrastructure, defining congestion pricing rules, provisioning loading zones in urban areas, and encouraging EV usage offer government a more active role in improving CC operations.

A quality threshold is a constraint in most model formulations based on shelf-life decay. However, quality-dependent demand is quite variable due to real-world consumer preferences, and this feature needs to be integrated with price optimisation. Progressively discounting items based on quality could be a meaningful way to reduce waste and extract more revenue. One would also need to incorporate game theory elements to prevent customers from exploiting such a system. This approach must work with waste reduction and disposal models to pursue globally optimal solutions.

Storage and transportation models in the literature include the impact of ambient temperatures. Real-life demand, too, depends on climate/weather (ice cream consumption, for instance). Nevertheless, this feature seems missing in the demand models based on price and quality parameters. Depending on when products might reach retail shelves and be available to customers, demand forecasts can be adjusted for expected weather, leading to improved distribution policies. This variation can be further integrated with consumer physiology and psychology data to understand demographic fluctuations and make suitable adjustments to the demand model.

Regular delivery of groceries to consumers as a subscription service (Wagner et al., 2021) balances demand and supply by reducing overproduction and lost sales. This business model must consider supply quality, shortages, discounting, recurring delivery schedules, optimal ordering, and pick-and-pack efficiency. Optimising such a service could strengthen the 'farm to fork' model for a just, healthy, environment-friendly food system, leading to sustainable agriculture and less waste.

### 6.2. Methodology

Sixty-three articles (some 27 % of the total) cover VRP and its variants in the context of CCs. However, detailed inspection reveals that most contributions relate to the solution method – novel *meta*-heuristics and hyperparameter tuning – with no significant innovation in model development. VRP methods are suitable for day-to-day route planning and multi-period planning, provided the locations and demand are constant. However, using them for tactical planning or integrating them with network design might be unrealistic, considering the variance in market demand (quantity and location) over time. Approximation methods may better address this uncertainty with exact solutions based on linear or non-linear programming models.

Grouping demand nodes using clustering methods is usually followed by approximating routing costs based on spatial demand distribution to model distribution costs. *K*-means is used as the standard clustering method in the articles reviewed. However, the algorithm-based approach is challenging to include in an all-encompassing optimisation model. A novel approach could be to model this grouping as a set-covering problem that can be solved using exact methods.

Most articles limit optimisation to one decision-making level (strategic, tactical, or operational). A global optimisation model that accounts for a higher demand variability at the strategic level followed by a lower demand variability at an operational one may be appropriate (based on scale and complexity) for achieving optimal solutions at each level. However, such approaches are missing in CC optimisation literature. Furthermore, the idea of feeding back implementation outcomes into the optimisation model parameters/constraints is not explored. For solutions to run with minimal analyst intervention, the ability to self-correct over time should be an integral part of optimisation models.

Driver familiarity with routes significantly impacts delivery performance and customer satisfaction. However, judging from our review, such knowledge is seldom considered when planning routes. Instead, this information could be incorporated as reinforcement learning in routing at a modelling level, facilitating the combination of machine learning and optimisation techniques. Since driving time impacts optimal CC distributions, such an approach could improve the effectiveness of route assignment to drivers.

In the articles reviewed, there is no validation of solutions by benchmarking against standard options. One remedy is deploying an open-source option providing VRP-specific solutions (e.g., Google OR-Tools) as the baseline for solution quality and performance. Improvement could then be quantified consistently across alternatives. Researchers commonly favoured the 'Solomon' dataset as a baseline for evaluation. However, based on demand points in Euclidean space, this dataset only offers 'random', 'clustered', and a combination of the two as options. Real geographies faced by LSPs rarely have such idealised parameters. It may be worth investing in building a dataset inspired by typical city layout models (e.g., concentric, grid, linear, sector) with demand influenced by a more realistic spread of population densities. Samples using such datasets will more likely resemble real-life scenarios with a range of road networks to benchmark the routing algorithms.

### 6.3. Technology

Delivery companies are experimenting with grocery delivery using autonomous vehicles in urban areas. Despite a low speed and limited load capacity, these vehicles can recharge themselves at docking points in the network, and it is easy to scale up this service (preference for contactless delivery might fuel this need). Optimisation models considering such vehicles' fixed and variable costs can be developed based on the VRP model for EVs proposed by Z. X. Zhao et al. (2020). Furthermore, drones could play a significant role in time-sensitive remote deliveries. However, drone-based distribution models must account for cooling/insulation needs and quality deterioration during the journey for temperature-sensitive goods such as medical supplies or perishable food.

The role of technology in CCs is ubiquitous. Real-time data from IoT (Internet of Things) sensors tracking warehouse conditions or container atmosphere allows for better visibility and decision-making. While the current literature focuses on real-time monitoring with sensor data,

using such data for real-time optimisation has not been explored. Integrating vehicle telematics with CC sensors can facilitate transport optimisation and real-time rerouting after vehicles, roads, and traffic incidents. We found no route optimisation algorithms or real-time rerouting decisions using this feature despite onboard navigation systems being the norm in delivery fleets. In storage, TTIs can be integrated with warehouse management systems to determine optimal 'slotting' on receipt, storage reallocation within the warehouse, and pick and pack decisions based on shelf-life information. Finally, linking vehicle IT systems with yard management systems of supply/demand nodes in distribution planning could reduce dwell time and deterioration of perishables.

A few articles refer to the role of cloud computing (Chen, 2020; Tang, 2022; Tsang et al., 2021), primarily facilitating faster parallel execution of complex heuristics for solving routing problems in CCs. The option of cloud-based solvers needs further exploration – using the cloud for solving optimisation problems could reduce local computing resource needs while scaling up hardware resources on demand. Additionally, applications can access multiple solvers that can be upgraded in real-time, choosing the best option per business needs. Implementing optimisation models using this approach requires further research to identify the best IT investment for a company.

### 6.4. Novel approaches

Urban transportation is one area impacted by the 'sharing economy' in the form of on-demand access to e-bikes, rental cars, and taxis. In CC transportation, allowing LSPs to access a pool of vehicles on-demand better balances demand and supply, especially when utilisation is spatiotemporally mismatched. While this leads to variability in pricing, algorithms matching demand and supply based on price ensure adequate availability of vehicles depending on priority. This approach optimises buy vs. lease vs. rent decisions in transportation assets. Also, pooling resources and tasks enables logistics teams to extend multi-temperature joint delivery efficiency from individual LSPs to the overall market. Of course, transport providers need to ensure that customer orders are duly marked to facilitate separation and that the mixing of different products is restricted based on business rules.

Delivery of online grocery orders includes chilled and frozen items, but challenges arise since many customers are unavailable when items are delivered. Companies are experimenting with insulated bags and limited access to customers' refrigerators with varying acceptance levels. However, the current literature (Leyerer et al., 2020) does not integrate the role of refrigerated parcel lockers in overall CC optimisation. These lockers present an opportunity to consolidate vehicle routes and unloading activities. Future studies could focus on cost-benefit analyses of locker investment and operations vs. delivery costs and customer satisfaction.

Refrigeration is a standard method to improve shelf life. A recent article by Giannakourou et al. (2020) explores an alternative approach, such as osmodehydrofreezing (ODF), to alter the deterioration curve for fruit and vegetables that may not suit freezing. Whereas ODF still involves eventual freezing, other options to change a product's state and later transform it back nearer the point of sale might not need cooling at all and thus yield net payoffs in savings vs. investments. For example, liquid milk can be converted to milk powder with a much longer shelf life and reconstituted on demand. The cost-benefit analysis of such options might provide opportunities to optimise operations and redesign SCs to support such transformation.

Typical temperature bands covered in the literature are chilled, frozen, and deep-freeze (usually below −18 °C). However, some vaccines need ultra-low temperatures (−60 °C to −80 °C). For example, while the need for ultra-cold storage for vaccines has been apparent since late 2020, no publications in our search address their CC optimisation needs. Another scenario is temperature control for chilled products when the ambient temperature is extremely low (e.g., Antarctic expeditions and, in winter, the Trans-Siberian railway), needing insulation to serve an opposite purpose so that the product does not suffer 'freeze damage'. Future research could focus on inventory, storage, and distribution policies addressing the unique challenges of managing ultra-low temperature CCs and CCs under ultra-low ambient temperatures.

## 7. Conclusion

Cold chains (CCs) control the temperature (by chilling, freezing, deep-freezing) of perishables such as medicines and foods, especially fruit and vegetables, which deteriorate over time at ambient temperature. Various supply chain (SC) players implement temperature control during processing, storage, and transportation. Optimising CCs can help improve SC efficiency (cutting costs and raising asset utilisation) and effectiveness (reducing waste and improving customer satisfaction). While existing reviews of CC papers cover technical and managerial aspects of these specialised SCs, ours uniquely synthesises the literature on optimisation models. We systematically reviewed 234 peer-reviewed journal articles in English on such models published up to and including 2023, the earliest detected in 2001. While adapting the Herfindahl-Hirschman Index reveals a disparate, multi-journal landscape led by Asia, especially China, an h-index of 46 measures the set's influence. Accelerated growth of publications (85 % postdate 2017 and 58 % postdate 2020) shows the field's rising importance, fuelled by global demand. That said, the paper has the limitation that articles might have used obscure terminology outside our search terms, occurred after our search cut-off date of 30th November 2024, or appeared in conference proceedings rather than journal articles.

We showcased critical components of optimisation models unique to CCs. Product decay as a function of time and temperature affects quality and shelf life, impacting demand and price. Fluid dynamics are crucial to cooling efficiency and effectiveness during storage and transportation. Cooling needs during transport influence fuel consumption and associated emissions, while product variety determines fleet characteristics (single- vs. multi-temperature vehicles). Time windows impact distribution routing. Finally, technology that monitors the overall CC (temperature, humidity, and gas composition) provides visibility and aids process control.

Using these components (temperature mostly being a constraint), CC optimisation models feature SC collaboration, strategic network design, storage design, tactical planning, and distribution. Non-linear product decay affects network design, storage, and distribution models, solved using linearising, non-linear optimisation solvers, or *meta*-heuristics. The packaging design of storage and transport containers uses the fluid dynamics of cooling. Decay and its impact on demand and pricing influence ordering/fulfilment policies and producer-distributor-retailer coordination. Most analytical models are transformed into ordinary differential equations with exact solutions.

Focusing on company case studies, research themes, and critical gaps suggested rich research opportunities. Better inclusion of the preferences and constraints of stakeholders such as consumers, retailers, government, and partners in a comprehensive model could enhance collaboration. For last-mile delivery in distribution, we suggested improved VRP solution methods and validation approaches. New technology in engineering (including bot/drone delivery) and IT (like cloud solutions) could lead the way in enabling CCs. We noted a systemic lack of replication/failure reporting, alignment with SC frameworks, and gaps around outage resilience and shelf-life transformation. Finally, we explored potential influences of social trends like the sharing economy, waste reduction, and end-to-end information visibility.

This first review of a niche but burgeoning field offers analysts and managers in SC operations state-of-the-art optimisation models applicable to CCs that could help benchmark current implementations and identify areas for improvement. Academics, practitioners, and policymakers will find insights into CC optimisation research opportunities or

potential optimisation initiatives. Challenges in global vaccine distribution will make CC optimisation even more relevant.

## CRediT authorship contribution statement

**Param Iyer:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Visualization, Writing – original draft. **David Robb:** Conceptualization, Resources, Supervision, Validation, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cie.2025.110972.

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